Depth, Oversmoothing, and Architecture: A Comparative Study of Graph Neural Networks on the Cora Citation Network

McMaster University

CSE 705 – Deep Learning

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Summer 2026

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2026-08-03

Abstract

[PLACEHOLDER: 150-250 word abstract summarizing the motivation (why depth and oversmoothing matter for GNNs), the comparative method (GCN, GraphSAGE, GAT at increasing depth on Cora), the oversmoothing diagnostics used (Dirichlet energy, MAD), the mitigations tested, and the key quantitative findings.]

Table of Contents

1 Introduction

[PLACEHOLDER: motivate the problem – why depth causes oversmoothing in message-passing GNNs, why this matters beyond a toy benchmark, and what this study contributes: a controlled comparison of GCN, GraphSAGE, and GAT across depth on Cora, paired with oversmoothing diagnostics and mitigation tests. State the research questions explicitly.]

2 Related Work

[PLACEHOLDER: prior work on (1) the GCN / GraphSAGE / GAT architectures themselves (PLACEHOLDER: cite PLACEHOLDER b, a, c), (2) the oversmoothing phenomenon and its theoretical characterization, and (3) mitigation strategies (residual/skip connections, PairNorm, JKNet, GCNII). Position this study relative to that work.]

3 Background

[PLACEHOLDER: brief self-contained background so the report does not assume the reader has the lecture notes at hand.]

3.1 Message-Passing GNNs

[PLACEHOLDER: unify GCN, GraphSAGE, and GAT under a common message-passing formulation; state where they differ (aggregation function, attention weighting, neighborhood sampling).]

3.2 Oversmoothing

[PLACEHOLDER: define oversmoothing as representation collapse with depth; state why it is expected to worsen with depth for these architectures.]

3.3 Oversmoothing Diagnostics

[PLACEHOLDER: define Dirichlet energy and mean average distance (MAD) precisely, including the formula, and explain what a collapsing value indicates.]

4 Methodology

[PLACEHOLDER: overview of the experimental design linking research questions to measurements.]

4.1 Dataset

[PLACEHOLDER: Cora citation network via PyG `Planetoid`, standard split, node/edge/ feature/class counts.]

4.2 Models

[PLACEHOLDER: GCN, GraphSAGE, GAT architecture details and the depth range studied.]

4.3 Mitigations

[PLACEHOLDER: residual/skip connections, PairNorm, JKNet, GCNII – as toggles on the configurable model class.]

4.4 Evaluation Protocol

[PLACEHOLDER: seeds, metrics (test accuracy, macro F1, per-layer Dirichlet energy, final MAD), and how results are aggregated (mean +/- std over 10 seeds).]

5 Experimental Setup

[PLACEHOLDER: software/hardware environment, hyperparameters (learning rate, weight decay, hidden dimension, dropout, epochs, early stopping), and how they were chosen.]

6 Results

[PLACEHOLDER: present results in the order: accuracy vs. depth per architecture (Figure 1), oversmoothing collapse curves (Figure 2), and the mitigation comparison (Table 1). State only what the numbers show; keep interpretation for the Discussion section.]

6.1 Accuracy Versus Depth

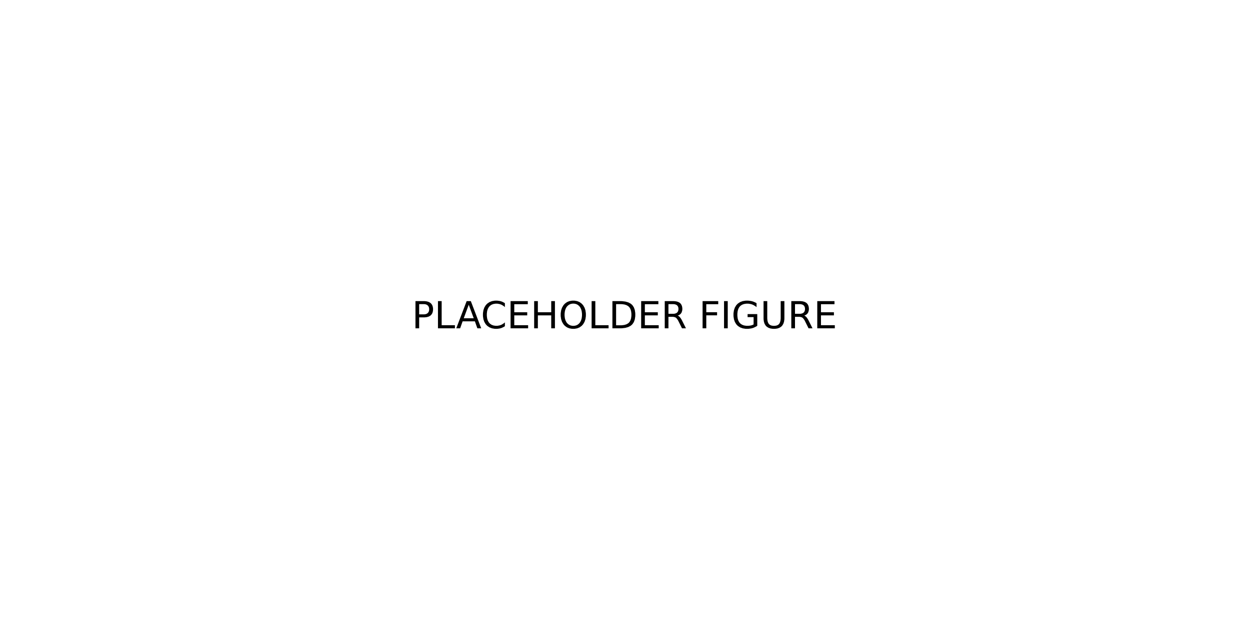

Figure 1: PLACEHOLDER caption – test accuracy versus number of layers, per architecture.

6.2 Oversmoothing Diagnostics Versus Depth

6.3 Mitigation Comparison

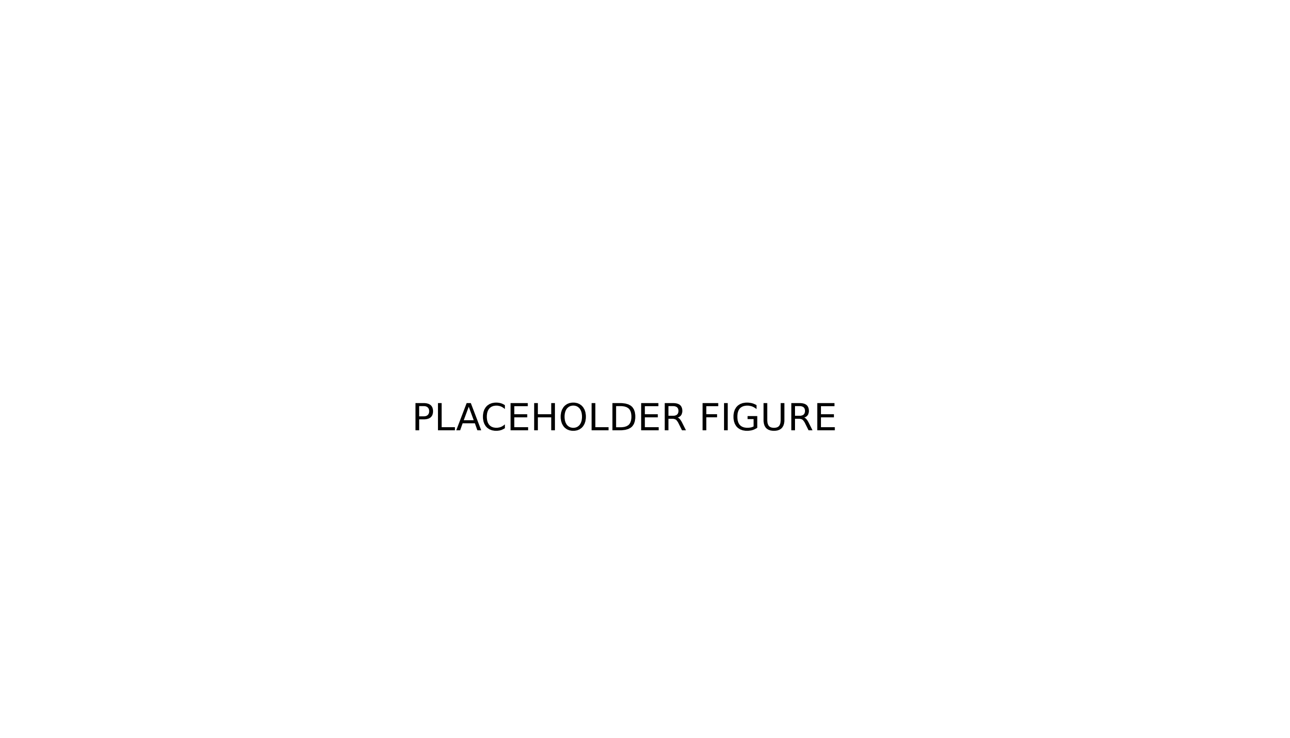

Figure 2: PLACEHOLDER caption – per-layer Dirichlet energy / MAD versus depth, per architecture and mitigation.

Table 1: PLACEHOLDER caption – test accuracy / macro F1 / final MAD by architecture, depth, and mitigation.

	architecture	num_layers	mitigation	test_acc	mad_final
0	PLACEHOLDER	PLACEHOLDER	PLACEHOLDER	PLACEHOLDER	PLACEHOLDER

7 Discussion

[PLACEHOLDER: interpret the results against the research questions – which architecture degrades fastest with depth, whether oversmoothing diagnostics track accuracy loss, which mitigations restore deep performance and at what cost. Note limitations.]

8 Conclusion and Future Work

[PLACEHOLDER: summarize the main findings and their significance; state concrete directions for future work, including the link to the physics-informed GNN (McPINN) thesis tooling if relevant.]

References

PLACEHOLDER, Author. a. *PLACEHOLDER – Replace with the Real GAT Citation.*

PLACEHOLDER, Author. b. *PLACEHOLDER – Replace with the Real GCN Citation.*

PLACEHOLDER, Author. c. *PLACEHOLDER – Replace with the Real GraphSAGE Citation.*

Appendix

[PLACEHOLDER: supplementary tables/figures not essential to the main narrative – e.g. full hyperparameter grids, per-seed raw results.]